

SAT Unit 1.3

Linear Equation in Two Variables

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Overview

- 1 Three Forms of Linear Equations
- 2 Transformation of a Linear Line
- 3 Special Lines
- 4 Real World Application

Introduction to Linear Equations in Two Variables

Definition: A linear equation in two variables is an equation that can be written in the form:

$$ax + by = c$$

where:

- a, b, c are real numbers.
- x and y are variables.
- a and b are not both zero.

Example:

$$2x + 3y = 6$$

is a linear equation because it has two variables and forms a straight line when graphed.

Three Forms of Linear Equations

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Standard Form of a Linear Equation

Standard Form:

$$Ax + By = C$$

- A, B, C are real numbers.
- A and B are not both zero.

Examples:

- $5x - 2y = 10$
- $3x + 4y = 8$
- $y = -2x + 7$ (This is not in standard form, but can be rewritten as $2x + y = 7$).

Question: Training Course Duration Difference

A certain apprentice has enrolled in **85** hours of training courses. The equation

$$10x + 15y = 85$$

represents this situation, where x is the number of on-site training courses and y is the number of online training courses this apprentice has enrolled in.

How many more hours does **each** online training course take than **each** on-site training course?

Answer Explanation: Training Course Duration Difference

Solution:

- The coefficient of x (on-site training) is **10**, meaning each on-site training course lasts 10 hours.
- The coefficient of y (online training) is **15**, meaning **each** online training course lasts 15 hours.
- The difference in duration per course:

$$15 - 10 = 5.$$

Final Answer: Each online training course takes **5** more hours than each on-site training course.

Understanding X-Intercept and Y-Intercept

Definition:

- The **x-intercept** is the point where the line crosses the x-axis. At this point, $y = 0$.
- The **y-intercept** is the point where the line crosses the y-axis. At this point, $x = 0$.

How to Find Intercepts:

- 1 To find the **x-intercept**, set $y = 0$ in the equation and solve for x .
- 2 To find the **y-intercept**, set $x = 0$ in the equation and solve for y .

Understanding X-Intercept and Y-Intercept

Example: Given the equation $3x - 2y = 6$

- **Find the x-intercept:** Set $y = 0$, solve for x .

$$3x - 2(0) = 6 \Rightarrow 3x = 6 \Rightarrow x = 2$$

The x-intercept is $(2, 0)$.

- **Find the y-intercept:** Set $x = 0$, solve for y .

$$3(0) - 2y = 6 \Rightarrow -2y = 6 \Rightarrow y = -3$$

The y-intercept is $(0, -3)$.

Question: Finding $\frac{b}{a}$

The graph of $7x + 2y = -31$ in the xy -plane has an x -intercept at $(a, 0)$ and a y -intercept at $(0, b)$, where a and b are constants.

What is the value of $\frac{b}{a}$?

Options:

A. $-\frac{7}{2}$

B. $-\frac{2}{7}$

C. $\frac{2}{7}$

D. $\frac{7}{2}$

Answer Explanation: Finding $\frac{b}{a}$

Solution:

- The x -intercept is found by setting $y = 0$ in the equation $7x + 2y = -31$:

$$7x + 2(0) = -31 \Rightarrow x = -\frac{31}{7}.$$

Thus, $a = -\frac{31}{7}$.

- The y -intercept is found by setting $x = 0$ in the equation:

$$7(0) + 2y = -31 \Rightarrow y = -\frac{31}{2}.$$

Thus, $b = -\frac{31}{2}$.

- Compute $\frac{b}{a}$:

$$\frac{b}{a} = \frac{-\frac{31}{2}}{-\frac{31}{7}} = \frac{31}{2} \times \frac{7}{31} = \frac{7}{2}.$$

Final Answer: $\boxed{\frac{7}{2}}.$

Slope-Intercept Form

Equation:

$$y = mx + b$$

where:

- m is the slope (rate of change).
- b is the y-intercept (where the line crosses the y-axis).

Example:

$$y = 2x + 3$$

- Slope = 2 (the line rises 2 units for every 1 unit increase in x).
- Y-intercept = 3 (crosses the y-axis at 3).

Point-Slope Form

Equation:

$$y - y_1 = m(x - x_1)$$

where:

- (x_1, y_1) is a given point on the line.
- m is the slope.

Example: Given a point $(2, 5)$ and slope $m = 3$, the equation is:

$$y - 5 = 3(x - 2)$$

which can be rewritten in slope-intercept form as:

$$y = 3x - 1$$

Transformation of a Linear Line

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Graphing a Linear Line

Graphing Steps:

- 1 Rewrite the equation in slope-intercept form ($y = mx + b$).
- 2 Identify the slope (m) and y-intercept (b).
- 3 Plot the y-intercept.
- 4 Use the slope to find another point.
- 5 Draw a straight line through the points.

Example:

$$2x + 3y = 6$$

Convert to slope-intercept form:

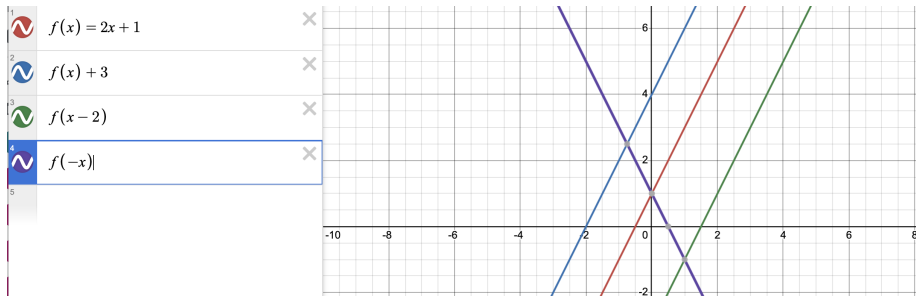
$$y = -\frac{2}{3}x + 2$$

- Slope (m) = $-\frac{2}{3}$
- Y-intercept (b) = 2

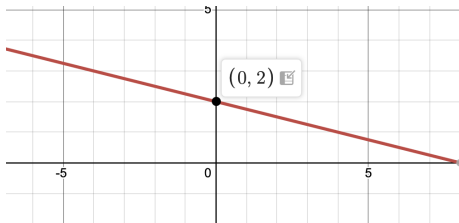
Transformation of a Linear Equation

Transformations of the linear function $y = 2x + 1$:

- **Vertical Shift:** $y = 2x + 4$ shifts the line up by 3 units.
- **Horizontal Shift:** $y = 2(x - 2) + 1$ shifts the line right by 2 units.
- **Reflection:** $y = -2x + 1$ reflects the line over the x-axis.
- **Change in Slope:** $y = 4x + 1$ steepens the line, increasing its rate of change.



Question: Identifying Function $f(x)$



The graph of $y = f(x) + 14$ is shown.

Which equation defines function f ?

Options:

- A. $f(x) = -\frac{1}{4}x - 12$
- B. $f(x) = -\frac{1}{4}x + 16$
- C. $f(x) = -\frac{1}{4}x + 2$
- D. $f(x) = -\frac{1}{4}x - 14$

Answer Explanation: Identifying Function $f(x)$

Solution:

- The given equation is $y = f(x) + 14$, so the graph represents $f(x)$ shifted upward by 14 units.
- Identify the equation of the given graph:

$$y = -\frac{1}{4}x + 2.$$

- Since $y = f(x) + 14$, we solve for $f(x)$:

$$f(x) = y - 14 = -\frac{1}{4}x + 2 - 14 = -\frac{1}{4}x - 12.$$

Final Answer: A.

Special Lines

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Parallel and Perpendicular Line Theorems

Parallel Line Theorem:

- Two non-vertical lines are **parallel** if and only if they have the **same slope**.
- Mathematically, if two lines have equations:

$$y = m_1x + b_1 \quad \text{and} \quad y = m_2x + b_2$$

then the lines are parallel if:

$$m_1 = m_2$$

Parallel and Perpendicular Line Theorems

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then the lines are parallel if:

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Perpendicular Line Theorem:

- Two non-vertical lines are **perpendicular** if and only if the product of their slopes is -1 .
- Mathematically, if two lines have slopes m_1 and m_2 , then they are perpendicular if:

$$m_1 \cdot m_2 = -1$$

Question: Finding the Value of d

In the xy -plane, line ℓ passes through the point $(0, 0)$ and is parallel to the line represented by the equation:

$$y = 8x + 2.$$

If line ℓ also passes through the point $(3, d)$, what is the value of d ?

Answer Explanation: Finding the Value of d

Solution:

- Since line ℓ is parallel to $y = 8x + 2$, it has the same slope $m = 8$.
- The equation of line ℓ , passing through $(0, 0)$, is:

$$y = 8x.$$

- Substitute $x = 3$ to find d :

$$d = 8(3) = 24.$$

Final Answer: 24.

Question: Finding the Slope of Line r

Line p is defined by the equation:

$$4y + 8x = 6.$$

Line r is perpendicular to line p in the xy -plane. What is the slope of line r ?

Answer Explanation: Finding the Slope of Line r

Solution:

- Rewrite the equation of line p in slope-intercept form:

$$4y = -8x + 6.$$

$$y = -2x + \frac{3}{2}.$$

- The slope of line p is $m_p = -2$.
- Since line r is perpendicular to line p , its slope is the negative reciprocal of -2 :

$$m_r = \frac{1}{2}.$$

Final Answer: $\boxed{\frac{1}{2}}.$

Real World Application

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Real-World Applications

Linear equations in two variables appear in various real-life scenarios:

- **Business:** Predicting profit and expenses.
- **Physics:** Calculating velocity and motion.
- **Economics:** Supply and demand relationships.
- **Engineering:** Designing bridges and buildings.

Example: A company's total cost equation is:

$$C = 50x + 200$$

where:

- C is the total cost.
- x is the number of products made.
- 50 is the cost per product.
- 200 is the fixed cost.

Question: Mixing Saline Solutions

How many liters of a 25% saline solution must be added to 3 liters of a 10% saline solution to obtain a 15% saline solution?

Solution

Solution:

- Let x be the amount of 25% saline solution to be added.
- The total salt in the solutions:

$$0.10(3) + 0.25(x) = 0.15(3 + x).$$

- Solve for x :

$$0.3 + 0.25x = 0.45 + 0.15x.$$

$$0.25x - 0.15x = 0.45 - 0.3.$$

$$0.10x = 0.15.$$

$$x = 1.5.$$

Final Answer: 1.5 liters.

Practices

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Question: Sticker Sheet Interpretation

A teacher printed sticker sheets with 7 stickers on each sheet. At a school party, she gave out 2 stickers to each student who attended. The relationship between the number of students who attended the party, x , and the number of sticker sheets that the teacher printed, y , is represented by the equation:

$$7y - 32 = 2x$$

What is the best interpretation of the number 32 in this context?

- A.** The teacher gave out a total of 32 stickers at the party.
- B.** The teacher had 32 sticker sheets remaining after the party.
- C.** The teacher printed 32 more stickers than she gave out at the party.
- D.** The teacher printed a total of 32 sticker sheets.

Answer: Sticker Sheet Interpretation

We are given:

$$7y - 32 = 2x$$

Here, $7y$ represents the total number of stickers printed (7 stickers per sheet $\times$ y sheets), and $2x$ represents the total number of stickers given out (2 stickers per student $\times$ x students).

So, $7y - 32 = 2x$ means the teacher gave out 32 fewer stickers than she printed.

Correct Interpretation: C. The teacher printed 32 more stickers than she gave out at the party.

Question: Perpendicular Line

Line m passes through the point $(-2, 4)$ and is **perpendicular** to the line:

$$y = \frac{1}{2}x - 3$$

Which of the following is the equation of line m ?

- A. $y = -2x$
- B. $y = -2x + 4$
- C. $y = 2x + 8$
- D. $y = -2x + 2$

Answer: Perpendicular Line

The original slope is $\frac{1}{2}$, so the perpendicular slope is the **negative reciprocal**:

$$m = -2$$

Use point-slope form with point $(-2, 4)$:

$$y - 4 = -2(x + 2) \Rightarrow y = -2x - 4 + 4 = -2x \Rightarrow \boxed{\text{A. } y = -2x}$$

Challenge: Relationship Between Two Lines

Lines L_1 and L_2 have the following equations:

$$L_1 : 3x + 4y = 12$$

$$L_2 : 4x - 3y = 5$$

Which of the following best describes the relationship between L_1 and L_2 ?

- A. The lines are parallel.
- B. The lines are perpendicular.
- C. The lines intersect but are neither perpendicular nor parallel.
- D. The lines are identical.

Answer: Line Relationship

Convert both to slope-intercept form:

Line 1:

$$3x + 4y = 12 \Rightarrow 4y = -3x + 12 \Rightarrow y = -\frac{3}{4}x + 3$$

Line 2:

$$4x - 3y = 5 \Rightarrow -3y = -4x + 5 \Rightarrow y = \frac{4}{3}x - \frac{5}{3}$$

Slopes: $m_1 = -\frac{3}{4}$, $m_2 = \frac{4}{3} \Rightarrow m_1 \cdot m_2 = -1$

Result: The lines are perpendicular.

Correct Answer: B.

Question

The table below shows the coordinates of three points on a line in the xy -plane, where k and n are constants. If the slope of the line is 2, what is the value of $k + n$?

x	y
3	7
k	11
12	n

Answer

Step 1. The slope of the line is 2, so:

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = 2$$

Step 2. Use points (3, 7) and (k, 11):

$$2 = \frac{11 - 7}{k - 3}$$

$$k = 5$$

Step 3. Use points (3, 7) and (12, n):

$$2 = \frac{n - 7}{12 - 3}$$

$$n = 25$$

Step 4. Then:

$$k + n = 5 + 25 = \boxed{30}$$

Question: Identifying Perpendicular Line Pairs

Consider the given system of equations:

$$5x + 7y = 1$$

$$ax + by = 1$$

where a and b are constants. The graph of these equations represents a pair of perpendicular lines.

Which of the following systems also represents a pair of perpendicular lines?

Options:

- **A:** $10x + 7y = 1$, $ax - 2by = 1$
- **B:** $10x + 7y = 1$, $ax + 2by = 1$
- **C:** $10x + 7y = 1$, $2ax + by = 1$
- **D:** $5x - 7y = 1$, $ax + by = 1$

Answer Explanation: Identifying Perpendicular Line Pairs

Correct Answer: B

Solution:

- Find the slope of the first equation by rewriting it in slope-intercept form:

$$5x + 7y = 1 \implies y = -\frac{5}{7}x + \frac{1}{7}.$$

The slope is $-\frac{5}{7}$.

- Since the lines are perpendicular, the second equation should have a slope that is the negative reciprocal of $-\frac{5}{7}$, which is $\frac{7}{5}$.
- Checking the options, the correct system with perpendicular slopes is:

$$10x + 7y = 1, \quad ax + 2by = 1.$$

This corresponds to:

B.

Summary

- A linear equation in two variables is written as $Ax + By = C$.
- The slope (m) determines the steepness of the line.
- Graphing requires finding intercepts or using the slope-intercept form.
- Systems of linear equations can have one solution, no solution, or infinitely many solutions.

Thank You!

We appreciate your time and focus.

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